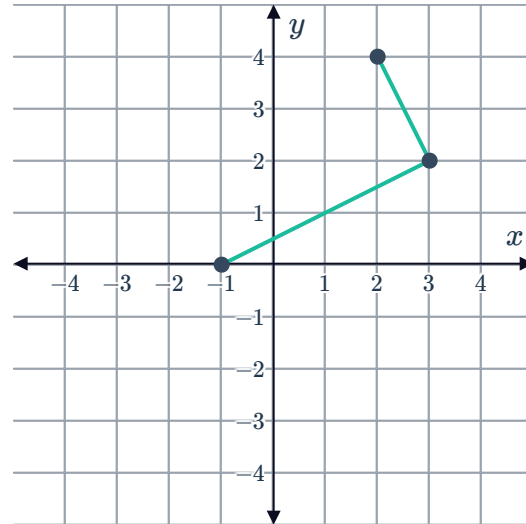


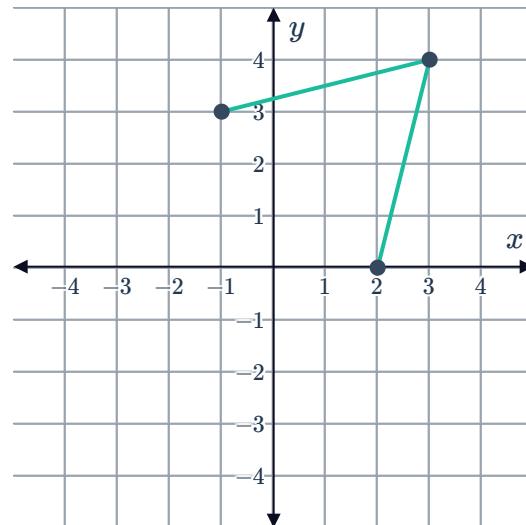
Coordinate methods with quadrilaterals



- 1 Consider the given diagram.
The points given represent three vertices of a parallelogram.
Determine the coordinates of the fourth vertex.

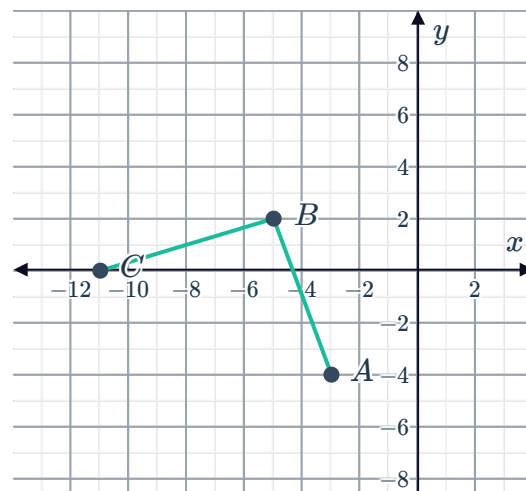


- 2 Consider the given diagram.
The points given represent three vertices of a rhombus.
Determine the coordinates of the fourth vertex if the missing point lies in the 3rd quadrant.



- 3 Consider the given diagram.
 $A(-3, -4)$, $B(-5, 2)$, $C(-11, 0)$, and $D(p, q)$ form the four vertices of square $ABCD$, where T is the midpoint of the diagonal $\overline{AC}$.

- a Find the coordinates of T .
- b Solve for p .
- c Solve for q .





4 Consider the quadrilateral shown.

a Find the slope of the following:

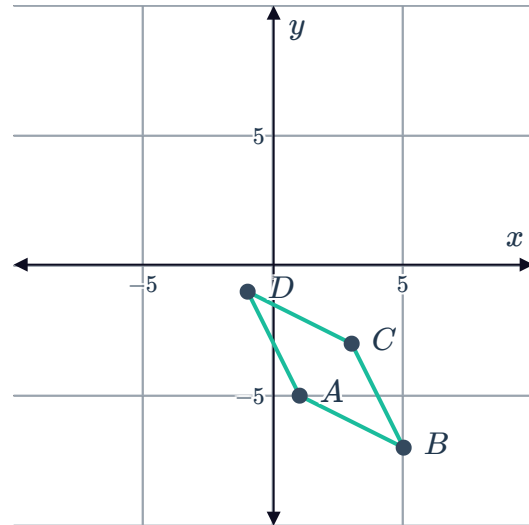
i $\overline{AD}$ ii $\overline{AB}$

iii $\overline{BC}$ iv $\overline{DC}$

b Find the length of the following sides:

i $\overline{AB}$ ii $\overline{AD}$

c Is this quadrilateral a rhombus or just a parallelogram?



5 The four points $A(-8, -4)$, $B(-5, 0)$, $C(0, -2)$ and $D(-3, -6)$ are the vertices of a quadrilateral.

a Plot the quadrilateral on a coordinate plane.

b Find the slope of the following:

i $\overline{AB}$ ii $\overline{BC}$ iii $\overline{CD}$ iv $\overline{DA}$

c Write down two pairs of segments that are parallel.

d Hence, classify the quadrilateral as precisely as possible.

6 Consider the quadrilateral shown.

a Find the slope of the following sides:

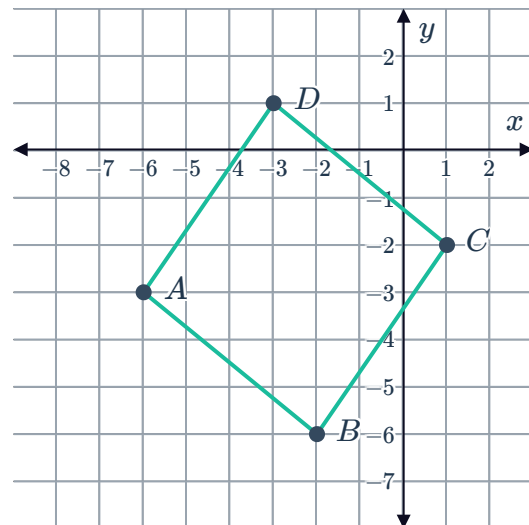
i $\overline{AB}$ ii $\overline{CD}$

iii $\overline{AD}$ iv $\overline{BC}$

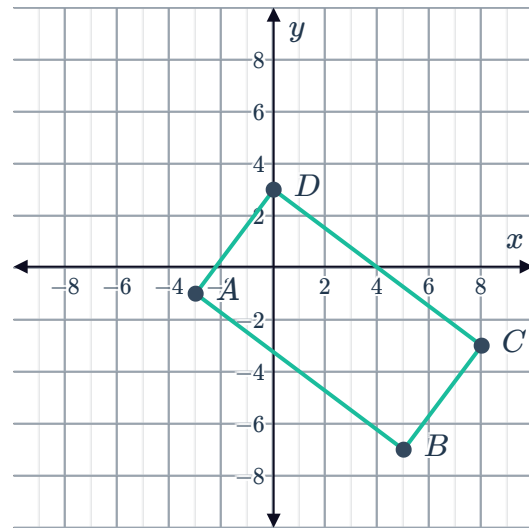
b Find the length of the following sides:

i $\overline{AB}$ ii $\overline{AD}$

c Hence, classify the quadrilateral as precisely as possible.



7 Consider the quadrilateral shown.



a Find the slope of the following sides:

- i $\overline{AB}$ ii $\overline{CD}$
iii $\overline{AD}$ iv $\overline{BC}$

b Find the length of the following sides:

- i $\overline{AB}$ ii $\overline{AD}$

c Hence, classify the quadrilateral as precisely as possible.

Additional questions

8 State the properties of each of the following triangles that can be identified with coordinate geometry:

- a Scalene triangles b Isosceles triangles
c Equilateral triangles d Right triangles

9 Consider the quadrilateral formed by the points $A(-3, -4)$, $B(5, -4)$, $C(5, 4)$ and $D(-3, 4)$:

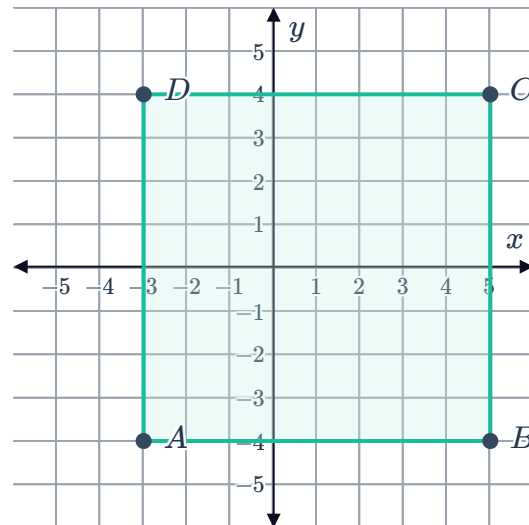
a Find the length of the following sides:

- i $\overline{AB}$ ii $\overline{BC}$ iii $\overline{CD}$ iv $\overline{DA}$

b State if all sides of the quadrilateral are congruent.

c State if all consecutive sides are perpendicular.

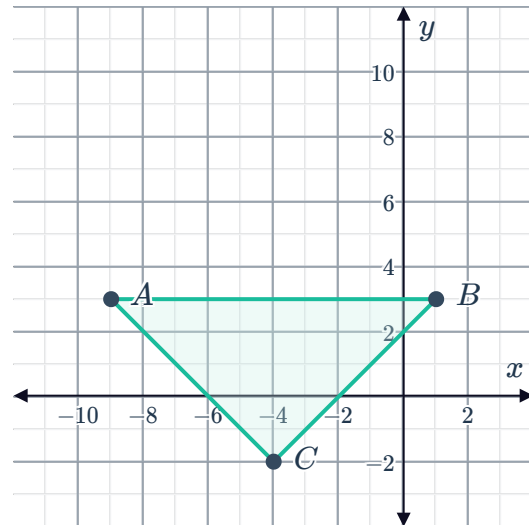
d Now, classify the quadrilateral as precisely as possible.



10 Consider the triangle shown below:

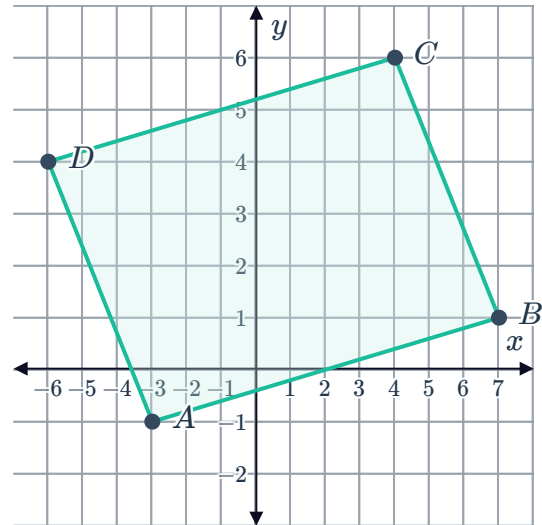


- a Find the slope of the following sides:
 - i $\overline{AC}$
 - ii $\overline{BC}$
- b Calculate the side lengths of all three sides.
- c Classify the triangle as precisely as possible.



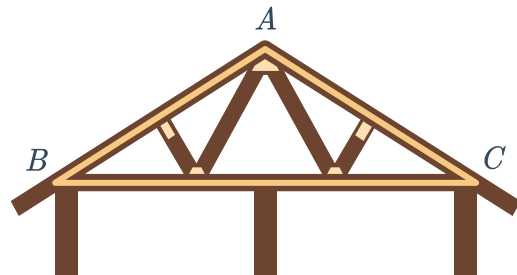
11 Consider the parallelogram $ABCD$:

- a Find the slope of the following sides:
 - i $\overline{AB}$
 - ii $\overline{CD}$
 - iii $\overline{AD}$
 - iv $\overline{BC}$
- b Classify this quadrilateral as precisely as possible.

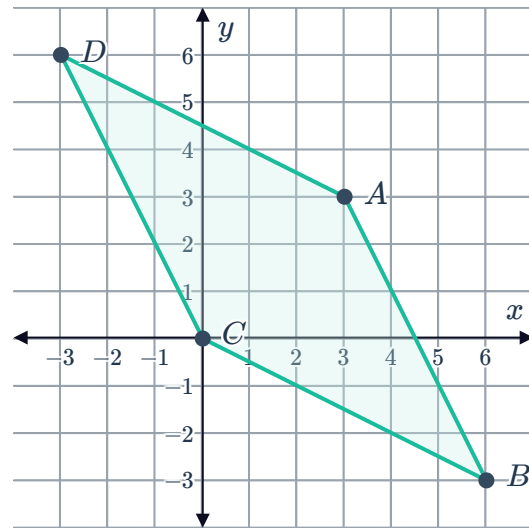


12 Isolda is designing a house using a graphing software. The outer pieces of the triangular roof trusses must form an isosceles triangle. The outer pieces of one of the triangular trusses is formed by three points: $A(2, 2)$, $B(-2, 0)$ and $C(6, 0)$.

- a Calculate the following lengths:
 - i BC
 - ii AB
 - iii AC
- b State if this is a possible roof truss.



- 13 Consider the quadrilateral $ABCD$:
Classify this quadrilateral as precisely as possible.



- 14 Consider a quadrilateral with vertices $A(3, 4)$, $B(6, 2)$, $C(5, -3)$, and $D(-4, 3)$:
- Plot the quadrilateral on a coordinate plane.
 - Classify the quadrilateral $ABCD$ as precisely as possible.
- 15 Consider a quadrilateral with vertices $P(4, -4)$, $Q(5, -7)$, $R(9, -11)$ and $S(0, -8)$.
Classify the quadrilateral $PQRS$ as precisely as possible.
- 16 A triangle consists of the points $A(-2, -2)$, $B(1, -4)$ and $C(5, 2)$.
Classify $\triangle ABC$ as precisely as possible.

- 17 Consider the following lines:

- Line P: $y = -6x - 4$
- Line Q: $y = \frac{x}{6} + 6$

- Line R: $y = -6x - 1$
- Line S: $y = \frac{x}{6} + 1$

- a Find the slope of each line:

i m_P

ii m_Q

iii m_R

iv m_S

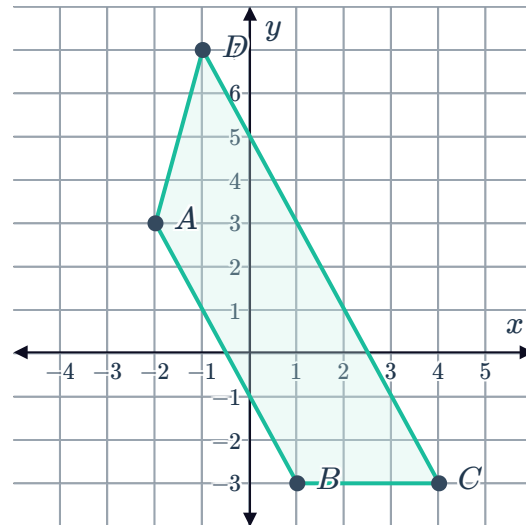
- b State the type of the quadrilateral enclosed by the four lines.

- 18 Quadrilateral $ABCD$ has vertices $A(1, -2)$, $B(9, 3)$, $C(7, -1)$ and $D(-1, -6)$:
Prove that $ABCD$ the diagonals bisect each other and therefore that $ABCD$ is a parallelogram.

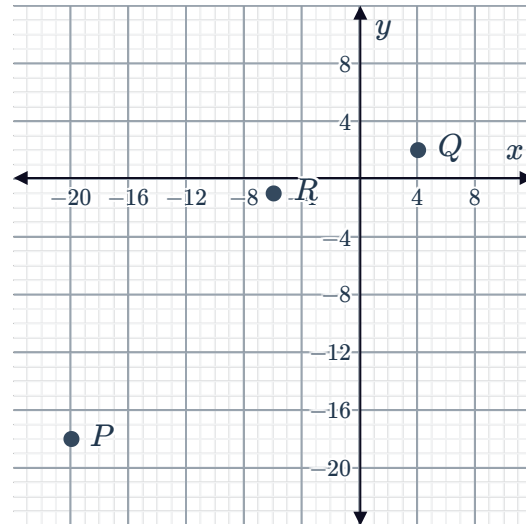
- 19 Quadrilateral $RSTU$ has vertices $R(3, 0)$, $S(5, -3)$, $T(-1, -7)$, $U(-3, -4)$.



- 20** Prove the the given quadrilateral is a trapezoid, but not an isosceles trapezoid.



- 21** Consider the points $P(-20, -18)$, $Q(4, 2)$ and $R(-6, -1)$:
Determine the coordinates of a point S
such that $PQRS$ is a parallelogram.



- 22** Describe two ways you can show whether a quadrilateral in the coordinate plane is a square.
Explain which method is more efficient.